

and Mouser allow engineers to select the height, width and depth they require so as to provide the closest dimensioned battery possible. The same is done while selecting solar panels as well.

Ratings and capacity. For selecting battery as a power source, attention must be given to its discharge capacity (1C and 2C) and energy rating such as (1000mAh or so).

For solar power, you can select small panels by considering their power rating, maximum power voltage, number of cells and other specs. Besides this, care must be taken in providing polarity proofing and recharging facility inside the device itself. Below is an example of battery selection for a microcontroller-based circuit that Knewron worked on, where 3.3V was the minimum necessary requirement and the device was to be run for 5 to 6 hours at one go.

"With approximate peak current consumption of 25mA, we chose a power source of 160mAh capacity delivered by a small LiPo battery that would meet these requirements. Due to its light-weight nature and small form factor, this LiPo battery was the correct choice," explains T. Anand, principal consultant, Knewron.

Terminals. Connectors are selected based on the need for a compact solution, or even for ensuring that the final device is tightly packed. Most off-the-shelf batteries come with different kinds of termination styles for battery connectors, such as PCB pins, pressure contacts or snap contacts to name just a few. However, certain applications might also require custom-designed battery connectors.

Some of the main factors affecting connector selection is surface-mount technology, compression style, through-hole interface to your PCB, different circuit sizes and battery pack orientation on the product.

Chemical composition. The designer will have to select battery technology based on its chemistry and also ensure that it does not have any substance of very high concern (SVHC).

An Innovative Photovoltaic Touch Panel Design

The US Patent and Trademark Office granted Apple, Inc. a rather interesting patent, which shows the innovative use of current technologies to enable a truly untethered device.

It takes the idea of a solar glass to a different level, where the photovoltaic panel is placed behind the glass (in this case a touchscreen display), so that the user can continue using his device while the solar panel absorbs energy and trickle charges the battery. Lithium-polymer batteries which are used inside these devices, are probably designed to withstand these constant-use cases and still deliver standard performance.

Optimising Your Power Source For Manufacturing

Designing for functionality is only half the problem solved, if its cost will balloon upon going into production. Getting a battery-powered device successfully to market requires the involvement of cost at different levels including design engineering, tooling, prototyping and production.

Designing depends on the type of application for which the battery is manufactured, therefore both high-power and low-power batteries are designed separately.

In high-power batteries, complex systems integration is required along with special safety considerations to prevent accidental damage and electrical and physical abuse. These high-power batteries are basically designed for the electrical vehicles (EVs) which force the designers to include both forced cooling and heating circuits.

On the contrary, low-power batteries require designing of precise-section plastic parts and the associated complex moulding tools. Most modern battery packs also have in-built battery management systems that require designing of integrated circuits (ICs) on a printed circuit board (PCB).

Most countries are very specific about Restriction of Hazardous Substances (RoHS) Directive and the EMI/EMC norms of the equipment. In such cases, even the certification stamp on the power supply/type of components used matters for the final design to be approved.

If we need a compact and portable battery, a Li-ion battery is chosen. These batteries are so common because they are some of the most energetic rechargeable batteries available and have a high energy-to-weight ratio. Comparing the different technologies is beyond the scope of the article, but we have featured two very new technologies targeting wearable electronics in the next section.

Battery technology for the Internet of Things

Internet of Things and wearable electronics both have one thing in common—unorthodox device form factors. In devices like watches, where cell batteries were traditionally used, the current generation of smart watches require something that can pack much more punch. With this in mind,

vendors have started coming out with batteries that allow designers to custom-build their own batteries for specific sizes.

Solcore is an embedded power solutions vendor, who has a range of very thin 3D printed batteries with lifetime recharging capability, customisable shape option and high energy density with no toxic materials inside.

This solves the problem of the shape, size and overall form factor for a portable electronic device. The lack of toxic materials makes it an easy choice for use in wearable electronics without attracting the ire of the final user.

Imprint Energy is another firm that enabled the creation of rechargeable zinc-based ultra-thin batteries without the safety concerns of other battery technologies. Considering the importance that manufacturers of wearable electronics place on user experience, this could be a must-have technology for the next generation of miniature electronics.

If you are using an ultra-low-power chip in your design, you stand to gain from the availability of computing chipsets that can function at a thermal